

ActInf Textbook Group ~ Cohort 5 ~ Session 20 (Chapter 9, Part 1) ~ 3/18/24

TextbookGroup/ParrPezzuloFriston2022/Cohort_5 | Cohort 5 ~ Session 20 (Chapter 9, Part 1) ~ 3/18/2024 | 1:00:58

all right welcome thanks for joining
um we're in cohort 5 in the first
discussion of chapter
nine in terms of the book we made our
way through the epistemic first half the
recipe for making a generative model in
chapter 6 then the discr and the
continuous time generative models in
seven and 8 now we get to chapter nine
which is the data driven component so
pretty much up to this point everything
has been like you think of it
conceptually you create the model and
then it generates synthetic
data that's using a generative model in
the direction that's usually called
generative it's generating data the
other direction of use for an active
inference model or statistical model is
as a recognition model so that taking in
empirical data and then fitting
parameters that recognize the patterns
in the empirical data so chapter nine
highlights that kind of back and
forth about designing generative models
that take in data that's being emitted
by empirical experiments like behavioral
experiments in lab so that's what
chapter N9 is about um you if you want
to ask the question that you had first
and then we can like jump in there
explore empirical data look at some pmdp
and empirical code just see what happens
but what what um was your question
before we

begin oh I I have no uh specific
question it's just I I noticed a lot of
people have commented on that paper so I
suspect it's
something completely a completely new
Direction in in the field so um
yes so could you could you comment
on the main point of this uh of of this
paper yes yeah great so this was the the
paper path integrals particular kinds
and strange things it was first uploaded
to Archive in
2022 and then it was um submitted to
This Journal called physics life reviews
so physics of Life reviews has a very
interesting um scope if you look at the
structure of this journal um they
publish like regular research as well as
comments on research and so during the
process of this paper being like kind of
undergoing peer review from the archive
version they also once they get to
basically the final version they solicit
from a pretty Broad and open Community
these short comment papers so like Ali
and I wrote a paper commenting on this
like just like a two-page paper those
comment papers are not peer reviewed
exactly they're more like editorially
overviewed um so I mean we see many many
names that we know because it's kind of
like an easy and fun way to to be part
of a bigger discussion get a citation
pump up the citations to the target
paper um so that's kind of the the The
Meta um on this okay so as for the
contents
itself um it is not entirely new it

builds on especially the 2019 fris and
work free energy principle for a
particular
physics um to connect it to the text
book all throughout the 2022 textbook
like we're talking about the continuous
time and the discrete time
models so um let's look at figure
4.3 so in the discrete time setting we
have a transition operator that
basically jumps you forward a discrete
Time Step In The Continuous time setting
the way that we've been discussing it in
the textbook is in terms of the
Taylor series approximation so starting
with a point evaluating at the point
then taking the first derivative around
the point and then the second derivative
around the point and so on um so that is
in a continuous
setting that idea of like doing active
inference in a continuous
setting is generalized and formalized
by Framing the free energy principle not
just in terms of the transition
probabilities in a discrete model not
just in terms of the Taylor series
approximation or the generalized State
space of continuous time model but
actually in a path integral formulation so
that's kind of like there's like a
thread that's like the likeliest thing
to happen that's the path of least
action like the baseball doing a
parabola like whether you knew about it
right at the beginning with the initial
conditions or whether you had tracked
the whole baseball it's like it does do

a parabola that's classical movement but still and then it's kind of like there's like a sheath around that Trace with paths that the baseball could have taken that are just exponentially less and less likely like it could have hit just a bunch of molecules in a certain way at this one point and kind of diverted off course but because the baseball is very large relative to air molecules it follows like a classical trajectory um so the first thing they do is they kind of reconsolidate the free energy principle in terms of ath integrals which can then be like dropped back down into the continuous time that we talk about in the textbook or the discreet time but like especially with maybe the ways that they're taking it even further like the path integral is kind of like the richest formalization like if you go back to the variational calculus or Origins with Richard fan they have to do with path integrals for particles so it's kind of like coming full circle in a way because the findan diagram was developed in the setting of of those exact questions now they're applying it to cognitive particles and then so that's one that's kind of like the consolidation of the formalism component and then the second major contribution and this is also piece I think that um Olie and I focused on in our commentary is like

previously people had always talked about well active inference can apply from kind of simpler inert systems to simpler active systems all the way on through complex systems and people had introduced like this notion of like mirror versus active inference for something that a mere active inference system is something you could model as doing active inference but it might be kind of simpler boring like yeah you could model that rock as flying but choosing not to fly but you've kind of brought out a big apparatus to model something that's not moving and so here with this figure and the discussion they brought a lot more clarity to these kind of classifications systems that don't evidence active States in an interesting way to a given Observer not to say they don't right map and territory then systems that have activity but the activity is kind of classical or conservative I don't know if a bacterium is really the Right image here I think something like a pendulum might be clearer because bacteria can also have Rich behavior all all the way on through what they want to highlight here which is like a metacognitive or a strange particle and that's also what um Lars and Lancelot highlighted in their comment was about the basian mechanics not just across the partition but okay do the partition and now in the internal States partition those again as if those

internal states have metacognition so it's it was a it was a big paper for consolidating um the formalism and for clarifying some of the different scopes I see yeah so so it's one of the top down approach to look at active inference yeah I I think so I mean it it doesn't start from modeling a particular system it starts from more axiomatic considerations so in that kind of like physics based free energy principle side whereas like active inference isn't mentioned in the abstract so it's like path integral formulation of the free energy principle systems that do that are active inference systems but here they're highlighting that that high road formulation see there there might be other pieces too and I think I mean the commentaries are I haven't read them all but they're probably people always do funny titles that suggest what they're about uhuh so maybe we can review them but in terms of getting diverse comments on a major piece of work this format which physics of Life reviews does and a few other journals is really

useful and like Carl and others have
used it multiple
times well um we can look at any part of
nine look at any
questions also I have um pmdp up in
cursor so we could explore with language
SL

models like how you incorporate
empirical data with the generative
models depends on whatever people want
to

do yeah I think looking at codes would
be

great okay

okay okay

[Music]

um okay we we'll we'll see what we can
do without um I I I didn't like prepare
anything specific so we might have to
debug a little bit or have a little fun
but

um Let's uh see

okay

um just just through context PDP
documentation and the uh package
itself infer actively SL pmdp so this I
just cloned in that repo so everything
that's in the GitHub I just pulled that
in um okay

so what do we want to look at or make a
model

of or is there an empirical data set
that we want to connect it to or a type
of empirical data that we want to
connect it

uh I I have nothing specific in
mind let's do a discrete Time
of we could we could try to recapitulate

one of the examples in the
textbook
like the L M listening to
music
from chapter
7 or if there's some other like
structure or type of data or kind of
action selection that people want to
explore we might be able to to do
it music is great
okay okay here just just
um
the there's a lot of ways to
um go go into this but um
okay let's start with
um let's just if this all runs then
it'll be good to
go okay so agents agent demo so let's
just see where where it is in the
documents
I just want to find one that already has
some
visualizations
um and then we can see if how how it's
built and how we can modify
it okay okay maybe the
Tas let's let's look at the teamas and
and go through it and then see where the
empirical data come into play but then
we can we could
like
um it doesn't have to be a spatial model
like maybe the um we we could see if
it's possible or like what would need to
happen to modify the teamas to be more
like a music example or
something I mean this is
um in in chapter 9 when they

overview here are the key figures then we'll go to the code first it's taking everything that was looked at before with a generative model and it's saying like Okay we were making a map of the rat in the te-as now let's zoom out to having the laboratory that encloses that rat in a teamas so we're choosing experimental stimuli and passing them to the rat those are given the label o because they're observations for the subject and then the actions of the subject are our observations in the lab setting so this kind of you you could have another layer outside of the laboratory too but this is just with focus on the the system of Interest this is like what the cognitive models enclosed in um 9.2 shows kind of the process of like kind of what you need to get for empirical results in a paper where you have a te-as build a PDP which we'll look at the code in a second then chapter nine is which I don't think will exist in the code so we can try building that part out um you go from empirical data and you kind of invert the model to run it backwards so that then you could test like five different rats and look at the statistics of how they differ five different treatments of of a drug so first just to kind of play with cursor I just said explain this notebook so right now for coding in active inference it it's like literally the most exciting it's ever been because

of tools Like This Way Way Beyond most
people's coding
potential at in a moment certainly far
beyond mine is the ability to actually
use cursor and other tools to to have a
conversation with the material so
there's a lot of questions that people
might be like kind of having burning
inside of them or like wanting to ask
another person that are very possible to
just ask through language models of the
code

itself here um is

importing the kind of General packages
that are needed and then specifically
importing the pieces of pmdp that are
needed that's

Imports

um here's the environment the
environment is going to be like where
the game or the setting is defined
we're we're in the discrete time
setting

um interestingly it's it's not a 100
like when when the setting is like right
arm wins it still is 98 and two so
there's a little
stochasticity even when the the context
is

like you know when when the it's on the
left arm or the right arm um maybe
that's just to keep keep it kind of a
little bit more in and
noisy um just if you want to
highlight the kind of action amidst
uncertainty then you know you could just
change this to one and zero but but um
keeping it this way highlights the

uncertainty part and here they call a

Tas environment

function searching the code base

describing what this teamas environment

call makes but it stores what's needed

for the teamas and puts it into

environment it's that method is in

another

file okay

here's going to be the hidden state to

outcome mapping that's the quote

collection of probabilistic

relationships called the a matrix this

is the mapping between hidden State and

observation that's a likelihood um

function and here there's there's

um we can

say output the

A and it will provide us but but this is

basically showing um

there's no

reward yes reward and there's there's

the the null State like with respect to

whether you're observing getting the

food or not let's just call it food um

but just call it reward because they

call it reward

um uh the top row ref The Columns

represent the locations that you could

be in the top row represents not tasting

anything the middle row

represents um tasting the the reward and

then the bottom row represents tasting

like the opposite of the reward like the

other arm so this is saying like the

starting and the Q location

hidden States map on to not tasting

anything the right maps on to tasting

the reward hidden State and then the left arm in this case maps on to tasting the opposite the Q mapping so we were in slice one of the a so the first kind of type of a matrix that comes into play is the one between location and taste there's also a second a which is going to be just a 2X two and this represents the mapping between the observed Q in the bottom of the te-as and what context you're in so in this setting it's kind of like an honest Q saying if you're if the um Q is saying that you're in the left better context it's it's accurate but you could flip that to make like a deceptive Q so that defines the two components of the a matrix now we're going to just go through literally a b c d b um is the transition Dynamics so here they start with kind of the simpler B which is going to be the transition dynamics of what context you're in so this one corresponds to the smaller 2x two a this is saying when you're in left better context as a true hidden state that stays that way whereas if it was like a situation where the context was shifting between time steps within a trial you would have an off diagonal now th those are uncontrollable Dynamics now they're going to get to controllable

transition Dynamics this is really the key piece in active inference because when all we have is the um downstairs part this is just a partially observable markov process that's just the A matrix and the B Matrix being inferred but there's no policy selection so to get into the policy selection that is equivalent to making oh to making B not just a transition Matrix but B has like multiple files or slices and then π selects choosing a policy choosing an affordance is equivalent to selecting which B Matrix is going to be applied to the continuation of the Hidden State transition so it's called controllable because there's a policy selection possibility for choosing which slice of B is applied whereas if you're just like listening to a song then you could infer the B Matrix but you wouldn't be able to choose the B Matrix

um there is um it's sometimes unintuitive the structure of these variables but again through printing them and asking you can get pretty far and they're they're they're formulaically laid out and also a lot of development effort like in the open and the closed Source spaces has to do with making the specification of generative models overall and especially the B Matrix a little bit more intuitive so that it can be like make it so

that this action does
that at this point you kind of have to
look at the Matrix and and encode how
that works so here this is like we're
looking at the um controllable
component of those two modalities so we
can't control the context left or right
better but we can control the location
so we're going from hidden state to
Hidden State at the next time step
there's four locations going to mapping
to the four
locations in this case because the
center is reachable from all
locations you can choose to move um to
the center from
anywhere similarly
they're just laying out these movement
options so those are the slices these
slice one two and three on
B those are the
affordances
okay they copy
in they copy in the variables they kind
of defined them just
um by themselves and now they're going
to copy them into this or GM
format it's not necessary that the
generative model is of vertical
representation of the generative process
at this point we were just defining A
and B like the vertical A and B but
we've just copied them in to be the part
of the agent's generative model but it
doesn't have to be that way you could
have the generative model of the agents
having a different structure than the
structure of like the map in the game

but that's just would take more code and everything

um now they're going to continue with the agent

class PDP is um like the major method is the agent class PDP is built to get to being able to build a discreet time agents as fast as possible essentially and make easy methods for calling that once it's there

so we're defining just zero is the controllable index because here

b0 the first b0o indexed is the controllable one but then the uncontrollable one it was B one so the second type of B

tensor this is just telling as we work towards building the agent that zero is the one that's under control whereas if it believes they control the context at the at the very best case you'd be wasting computational resources on Computing that

um here the agent is defined in terms of just the a the B and um the control

indices now it gets defaults for everything else and those are going to be kind of like investigated a little bit it starts out with a flat

prior for the hidden state it starts out with a flat so it has it it doesn't know where it is in the beginning it has an equal belief across the four possibilities of

location and that's plotting d0 corresponding to location that's the controllable index so d0 corresponds to the prior on hidden State B 0

corresponds to Transitions of hidden
State a z corresponds to Hidden State
and observations it's like and then the
context is that second um
modality and so here it has a um even
prior on which contexted
it then they modify it so that it has a
precise and accurate belief one hot just
means a variable with a one in one
location and then the rest of them are
zeros so they've modified that to now um
look like a strong belief that it's in
the zero
position um okay now they get to see so
after having done this they will have
defined a b c and d then that will have
basically constructed the agent so
they're going to um go to that um reward
modality there's some interesting
discussion about like how do you
actually connect up like an observation
to to be
incentivizing and and here's their um
here here's the C preference
variable okay that was it defining the
team as ABCD of the agents here's the
active inference
Loop
so let's let's do something
like doesn't always work on the one shot
but a shocking percentage of the time it
does fine
it literally did it it literally did
it so here's three or here's four time
steps it's funny you know it's the half
time steps obviously don't matter so
here it's
um these plots

represent that octave inference
simulation so it's
like that's always what's super
interesting is defining the agent like
the kind of scaffold or the template of
the agent is really just the
beginning because then there's a ton of
methods to describe like to visualize
the outputs of the
agent and to sweep across different
parameter
combinations
okay they're kind of now
plotting
um they're kind of inspecting the
beliefs plotting the updated
beliefs and then earlier I was just I I
I I didn't really make any but just
started to visualize a few
things
okay let's copy this
again
so here we'll do the the chapter nine
thing and bring in empirical
data it's just some some it's doing a
few just
minor um just format adding but these
lines are not being changed but green
means it's adding in a line red it's
leeting
you yeah uh I I want to ask about the
the gp4 does it actually read the paper
and the documentation why I this
understand yeah good good question like
in this
case um well first off if if you were
just using like the chat interface for
GPT or like whatever Cloud 3 Etc um you

could just copy in code and then say
like improve this script and then it
would you know based upon the statistics
of all the code it had seen it would
update that
script user takes that a lot
further because
um it does code base in
indexing
so when you type in a prompt to GPT
first it tokenizes your input converts
the words into tokens then those tokens
are um calculated in terms of their
embeddings in the semantic
spaces then like that embedding can be
used to then generate more things in
that semantic neighborhood
shortly the cheap part of using a trans
formal model is actually going from the
tokens to the
embeddings whereas going from the
embeddings to generating new synthetic
material is costly and then obviously
fine-tuning it and training it is even
more costly so what cursor does is in
the folder that you're
in it it indexes the
code so it runs all the code through the
first part to embed it and then when you
do a
Quest it like
um so control
enter so here it's scanning across it's
here GPT is going to be tokenizing and
then checking the embeddings of this
prompt I wrote then it uses basically
what's called retrieval augmented
generation rag you might be using that

in some other variants to find which sections in the code itself actually have relevant information in terms of having a similar embedding so like if you had PDFs of a bunch of different disciplines and then you said well like what reacts with methane and then it would look for where methane had been introduced or methane is similar to butane

if that if there was a paper that said methane is like butane then it would to a lesser extent draw on papers that talked about

butane so this is not co-pilot no you can use co-pilot as well in this setting but this is cursor. sh cursor do okay got it thank you this is yeah it's a vs fork and and and even if you've been away from it for like a few days you should download a new version of it because like they're the the changing is very rapid

so okay let's let's see how well this did here's the here's what it wrote for visualizing a team as she it's like it didn't output anything I'm not seeing anything

so there's there's the side chat mode where um you can chat with the codebase and then there's kind of this inline editing mode and there's also kind of a tab autocomplete mode that's more like the co-pilot but this inline version is

there's probably so many in its training
data set so many simple code examples
like just checking if there's a folder
like this kind of
logic except the changes

let's

see team a

e

so that's kind of the danger sometimes
to to an extent is like it's really easy
sometimes like be spinning your
wheels with with code that can be like
kind of overwhelming and look relevant
and the code might not be like bad by
itself but then like you start stacking
all this code and then like the
maintenance and interpretability costs
can be very high

he

like it's fun also just like this is
where we're at I mean this is a chapter
nine you know this is kind of where
we're

at as we continue and other people
continue to first off ask GPT about
active

inference and it includes more and more
recent checkpoints in its training and
of course web enabled search like
perplexity and all this like it gets
better at doing active inference second
huge piece is going to be
um more and
more actual examples of active inference
code especially for like higher
reliability

settings it's cool that there are PDP

demos at the same time like I know that

the community has more than just a couple notebooks but they're not in the pmdp repo so then the cursor only has a of examples to go off of so adding more working examples updated language models with internet access and better code scoping um fine-tuning language models on transcripts from the active inference journal from like our live streams and model streams and the textbook group questions that people ask getting folded back in to fine-tuning and to Rag and model scope if we pulled in the Journal transcripts for all the live streams then we could say something like um make it into continuous time and then maybe in several discussions somebody said like well what's the difference between discreet and continuous time and then that was addressed and then it could go off of those questions to actually inform it so especially for a field like active inference where it's like this is the domain expertise so that being invoked will help a lot but just for doing things like improve and generalize and professionalize this code uh so to pulling data from active Journal does it require tuning like we're not totally set up to do it today but yeah in principle you could pull in the the GitHub repository

for the octave inference
journal and then it would have the
transcripts in the plain text but there
might be a bunch of redundant stuff like
we might have four different versions of
the plain text or different translations
or something like that but yeah we could
we would prepare a kind of rendering
flag like output a single version of
every transcript rpt with no time steps
and have
that and then and also have papers
and more code examples and all of
this so like even just playing around
with where it's
at helps
calibrate
um we we can say generate
so with these kinds of
functions like here it this is not
um it's not outputting the data exactly
like we need it to
so even just playing with these like
it's it's really some of the most and
fastest
fun I've ever had using active
inference because um as long as I yeah
go ahead so as as I don't uh exceed the
context length I can just put in
whatever repo I
want actually you can pull in repos that
are Way Beyond the context length so
let's just say that the context length
was 100 just just make it simple and you
pulled in um a repository that was a
thousand so the first thing that would
happen is the Thousand lines of code
would be embedded that's the syncing of

the embeddings then then let's just say you did a 20 um a prompt of length 20 so you've used up 20 of the 100 of the context window with your prompt then that is going to get embedded and dispatched to cursor and then cursor is going to do its best to fill the other 80 of the context window with the most relevant 80 from the code base that's this part where it's scanning the codebase and then pulling in in the context so then the final execution from cursor is your 20 prompt plus the 80 most relevant that rag obtains so it may pull in like misleading examples but to the extent that that that the the macros and the patterns do exist in the code base it's basically effective then the less that what you're asking for exists in the code base the less of course it can rely on it and so then the more it's just relying on the overall statistics of the language model another thing is that um cursor you you can um you can set um I I don't know where it moved the setting to but you can force it to call another you can force it to call a different endpoint for the language model so then you can use like jan. a or LM studio. a or another one to have a local language model so then cursor is dispatching to a local language model and these are also including code

enabled and so
on so it's like kind of interesting
because it's like it's not the math of
but as that gets brought in and then for
the equations like we already have every
equation
um in the latc
form so this is highly
readable by the
models many can also look at images too
but what it's going to do when it gets
an image is basically OCR and try to
reverse to latch anyway so it's
enormously costly so there could be some
repos like maybe the textbook group repo
that just have like all the equations in
latch or the whole text in plain text
stripped of some you know um just spaces
and kind of cleaned up a little bit and
then those kinds of those kinds of plain
text files
can be brought in with one line in in
git and then you functionally would have
material because it's not it's not only
looking at code when it's doing its
scoping like here it it looked into just
plain
text
so you know Long Live digital stigma gy
contributions and questions that people
make really do
matter even if they're a starting place
for being improved and
corrected it it really it really does
have compounding
effects on the total active EOS
system like we could we could have a
function to generate synthetic teamas

data so that might be one step would be
okay we made the team's generative model
now let's make a generate synthetic data
and then let's do chapter nine kind of
get this pairing between the generated
synthetic data and the the generative
model then you build another function to
go from video
data to the same format as the synthetic
data but you already established that
that synthetic data format was able to
be parametrically useful for the
GM and you could have already developed
the visualization sweeping other
methods so it can be quite a large code
base and you can you can write like
thousands of lines in a
day because it it just spills out these
functions so I think a lot of the
previous discussions in chapter 9 in the
textbook like they're they're either
philosophical or analytical I mean or
both and um yet the
applying is very much
like Computer Engineering and
statistics and as the models and the
programs get
like in 6 months I think a lot of these
will be even
better uh sorry one more
question uh agent based modeling uh I
wonder if that's a field suitable for
inference yes I mean that's why pmdp the
central type is the
agent active inference is Agent based
agent based modeling whether you do it
at net logo or any other like
kind has always had like the kind of

incoming perceptual and outgoing action
selection so that kind of cybernetic
framing of single and multi-agent
ecosystems of shared intelligence is not
new to active
inference really the essential piece is
that that entire
generative model of the agent is being
understood in a unified way right memory
attention etc etc etc all the diverse
cognitive phenomena are
being approached in a unified Manner and
with a unified loss or fitting function
free energy under the free energy
principle but you'll find other agent
based models that have a perception and
action
component
uh if I just look at net logo uh it's
not explicitly uh done in in the
language of active
inference right so I I wonder if there
is value to be added if you if you bring
the active inference discipline into
that
field yes
certainly yes yeah like going to the
classic net logo simulations re wrting
them to be more active inference and
like it's a two-way street people in net
logo then can compare active inference
you can make um portfolios or models
that have actm and non- ACM agents in
net
logo and then it's instead of like oh
well you need to learn this like other
software it just brings active inference
into whatever software is being used and

it might be like as simple as just using
a different loss
function so my I think oh sorry yeah
yeah sorry uh okay I just finished my
question I wonder if that's a good uh
topic for a master
thesis uh I I I think it could be I mean
like uh in terms of the implementations
we have I've never seen a net log
go so it could be it could be
interesting you know something else that
that U my adviser works on is mut Mutant
agents and I haven't really seen
anything in the literature so far on
active inference about
mutation like between
Generations
yes yeah that could be really
interesting like just lock in the schema
and the understanding for the teams and
then do a lifetime where it does 100
teams and that's the fitness function
and then have Rec
combination I mean I I I I don't I'm not
set up to Run net logo exactly right now
but this looks like net
logo so this just gets super far this
gets like past
80% in a few minutes in terms of getting
somewhere so that's that's Mega
exciting for for a lot of
reasons butan sorry did did Jeff yeah
what uh did Jeff mention mutation
agents
yeah is there a paper
somewhere yes I can link it give me a
second cool but yeah f um fun times
interesting chapter n um

discussions one interesting meta repo
would be if we made a repo that just I
mean like literally I may do this
tomorrow just brings in all these other
repos but then at that point it might
start like pulling context from like
across repositories and it's not useful
again so there's a lot of
metaprogramming work and documentation
that will make it more accessible and
useful and and and vastly more
unexplored than
explored let see there's the paper cool
yeah
thanks sometimes I like like just
just cing
through okay so it's on the screen for
you know like two frames at you know for
a tenth of a second or half half a
second or whatever it
is future
screenshotter will do
better cool all
right that
was the first discussion on chapter n
yep
Susan uh yeah yeah I'm my brain's you
going 100 miles a minute and so the the
existing models that's in there are
um where would you say the level of
sophistication um what what would be the
next level of
sophistication that's not there yet in
PDP yep in terms of yeah the the
artifacts that are are are there now in
terms of
of going yeah get going up into levels
of abstraction for example um modeling

um agents

negotiating um you know which I'm

assuming you know we're not even close

to there yet but um

but so beyond the yeah the rat of the

maze what do you what do you see will be

the next step in evolution of the

models yeah like currently the main PDP

package only has these several

documentations and simple examples um

like I know there's a THX more work not

in the repo so I would like to see the

next consolidating step to be many many

diverse success uccessful simple

examples and then the introduction of

motifs like metac cognition and other

things like that and then that provides

like the pallets to be recomposed into

these more composite simulations so

we're in the game it's all happening yep

and saying and asking and writing is

more than enough so thank you all for

this fun time see you next time thank

you by see you next bye